

Sivangi Chatterjee

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EDUCATION

University of Pennsylvania

Aug 2025 - May 2027 (expected)

Master's in Computer and Information Science (MSE CIS); GPA: 3.57/4

Philadelphia, PA

Relevant Coursework: Software Systems, Machine Learning, Principles of Deep Learning, Database and Information Systems

Maulana Abul Kalam Azad University of Technology

Aug 2018 - Jun 2022

BTech (Hons.) in Information Technology; GPA: 3.87/4

Kolkata, India

Relevant Coursework: Mathematics-I&II&III (Calculus, Linear Algebra, Probability & Statistics, Differential Calculus), Data Structures & Algorithms, Operating System, Artificial Intelligence, Cyber Security

TECHNICAL SKILLS

Programming & OS: C | C++ | C# | Java | Python | HTML | CSS | JavaScript | TypeScript | SQL | Linux | Unix | Windows

Frameworks and Libraries: Spring | Spring Boot | React | Node.js | Spark | Microsoft D365 CE | PowerApps | ASP.NET

Machine Learning: Natural Language Processing (NLP) | LLMs | PyTorch | Keras | Scikit-learn | NumPy | Pandas | HuggingFace

Database & Cloud: OracleSQL | Postgres | MySQL | MongoDB | Neo4j | Azure | AWS | Docker

WORK EXPERIENCES & INTERNSHIPS

Lexmark | Xerox (India)

Jul 2022 – Jul 2025

Software Engineer, IT Applications

Kolkata, India

- Designed and delivered a centralized supply chain forecasting module utilizing **Azure**, **.NET**, and **Dynamics 365**, automating inventory planning to drive a **\$10M reduction** in inventory overhead while restoring pre-COVID back-order levels.
- Led the end-to-end migration of legacy customer support systems from **Siebel to Dynamics 365** utilizing **Azure** and **.NET 8**, accelerating workflows to achieve a **50% productivity boost** and a **20% increase** in customer satisfaction.
- Built and integrated Managed Print Services (MPS) and enterprise hardware modules into the core sales platform leveraging **JavaScript**, **TypeScript**, and **.NET** plugins, expanding the product ecosystem to drive a **30% revenue increase**.
- Collaborated with product managers and QA in a SAFe Agile environment to define feature requirements, resolve bugs, perform code reviews and draft technical specifications across **12 PIs**, thereby boosting system efficiency and user satisfaction.

Nomura Research Institute Financial Technologies India Private Limited

Sep 2021 – May 2022

Associate Software Engineer Intern

Kolkata, India

- Enhanced configuration management for security brokerage platforms using Java and Oracle, decreasing deployment time by **20%**.
- Boosted data accuracy by **15%** using field, text, and substring matching algorithms, CLI-based validation tools, and by resolving client-reported discrepancies with a focus on critical fixes and client-centric issue resolution.

RESEARCH EXPERIENCE

Cognitive Computation Group, University of Pennsylvania

Jan 2026 - Present

Research Assistant, Advisor: Prof. Dan Roth | Mentor: Haoyu Wang PhD

- Investigating debugging and self-correction behaviors in LLM-based agentic pipelines on multi-step tasks (e.g., SWE-bench-style settings)
- Constructing trajectory-level datasets by perturbing intermediate agent steps to study and improve error recovery via fine-tuning.

APPLICATION & RESEARCH PROJECTS

Generative Social Persona Modeling & Simulation

Nov 2025 - Jan 2026

Machine Learning Research Project

- Developed a generative simulation framework across **25 personas** to synthesize data for **400:1** imbalance and enable open-ended responses.
- Benchmarked Random Forest/MLP on **523k** records by fusing **384-dim all-MiniLM-L6-v2** embeddings with structural metadata.
- Fine-tuned **Qwen2.5-3B** via **LoRA** ($r = 16$) achieving **63.2% Accuracy** and **0.81 AUC**, a **30%** lift over zero-shot baselines.

Bidirectional Co-Attention for Image-Text Retrieval

Nov 2025 - Dec 2025

Deep Learning Research Project

- Engineered Bidirectional Co-Attention fusing frozen **ViT-B/16** and **BERT** features for fine-grained, pre-pool mutual refinement.
- Achieved **51.0% R@1** on Flickr30k via symmetric InfoNCE, surpassing CLIP-style models by **20%** and cross-attention by **66%**.
- Resolved modality heterogeneity via mutual attention across **197** visual tokens, preventing pooling loss and boosting alignment by **8.5%**.

Job Market Analytics & Intelligence Engine

Oct 2025 - Nov 2025

Full Stack Development Project

- Built frontend in **React** and backend in **Node.js**, aggregating **184k+** records via **REST APIs** on AWS RDS to visualize hiring hotspots.
- Engineered a **3NF PostgreSQL** schema enforcing **100%** integrity via delayed-constraint **ETL** to manage many-to-many relations.
- Optimized SQL latency by **70.7%** (3.4x) via **covering indexes** and window functions, reducing analytics execution to **0.8s**.

PUBLICATIONS AND PREPRINT

Audio Processing using Pattern Recognition for Music Genre Classification – Sivangi Chatterjee, Srishti Ganguly, Avik Bose, Hrithik Raj Prasad, Arijit Ghosal  *preprint* [Preparing for submission to BVICAM's International Journal of Information Technology.]

ACHIEVEMENTS

Manager Appreciation Award | Lexmark International (India) Pvt. Limited

Recognized for exceptional contributions to the PCM Shadowing project (part of MVP Phase 3) while effectively addressing resource challenges.